

GUJARAT POLICE SENTINEL HACKATHON 2026

159 / 159 TESTS PASSING

DECLARED: HYBRID MODEL

PRAHARI (પ્રહરી)

Integrated Video Management & Analytics Platform across 26 Independent
Government Departments

80,000
CAMERAS STATEWIDE

0.45G
PEAK WAN GBPS

₹130
Cr
5-YR TOTAL COST

Repository: [AJ-EN/prahari](#) · Production Codebase in Python, ONNX & SQLite/PostGIS

✗ The Central Streaming Trap (Model 4)

Streaming 80,000 cameras to Gandhinagar breaks under the laws of network physics:

- **160 Gbps Continuous WAN Load:** Requires massive dedicated telco lease lines costing ₹60+ Cr every year.
- **52 Petabytes Central Storage / Month:** Astronomical SAN/NAS storage footprint for raw, mostly inactive video.
- **Single Point of Failure:** A single backhaul cut blinds central police dispatchers statewide.

✓ The PRAHARI Law (Federated Edge)

Video stays where it is born. Only observations cross the state.

- **0.21 Gbps Avg / 0.45 Gbps Peak WAN:** Over **~360× reduction** in peak statewide network load!
- **~210 TB Central Storage:** Only lightweight text events (~550 B) and forensic crops (~15 KB) are transmitted.
- **Autonomous Island Mode:** Edge nodes continue local inference during network cuts; zero event loss.

160 Gbps → Under 0.45 Gbps at Peak (~360× Reduction)

EMPIRICALLY DERIVED FROM MEASURED CAMERA BITRATES AND PEAK TRAFFIC K-FACTORS

26 Siloed Departments

Police (Home), Food & Civil Supplies (Godowns), RTO (Tracks & Tolls), Health (Hospitals), GSRTC (Bus Stations), Urban Municipalities.

- Independent VMS platforms (Milestone, Genetec, Dahua, Hikvision)
- Separate 5-year AMC maintenance contracts
- Conflicting retention policies (7 to 30 days)

1,000 km Dispersion

CCTV assets spread from border outposts in Kutch & Banaskantha to industrial corridors in Valsad & Surat.

- Extreme network variance: 1 Gbps fiber to 2G/4G cellular links
- Remote tribal blocks (Dahod, Dangs) with frequent WAN cuts
- Latency spans 5 ms to 450 ms across the state

Extreme Heterogeneity

Massive diversity in physical hardware and video compression formats.

- Analog DVRs alongside modern 4K IP cameras
- Mixed codecs: H.264, H.265, and MJPEG
- Non-standard RTSP implementations and fluctuating FPS

Strategic Takeaway: Any solution requiring departments to replace NVRs, change public IPs, or void AMC contracts will stall in inter-departmental deadlock. **PRAHARI** requires zero hardware changes.

REFERENCE MODEL	ROLE IN PRAHARI	TECHNICAL IMPLEMENTATION	ARCHITECTURAL JUSTIFICATION
Model 1: Registry & GIS	MANDATORY FOUNDATION	PostgreSQL + PostGIS Central Database	Unified visibility layer, 3 onboarding paths, automated coverage gap analysis.
Model 3: VMS Federation	FEDERATION SPINE	Schema-driven Event Bus & Middleware	Integrates existing Milestone/Genetec/Hikvision NVRs without disturbing local operations.
Model 2: Direct Ingest	EDGE VIEWING LAYER	RTSP / ONVIF Direct Adapters	Connects standalone cameras lacking local VMS hosts into the unified wall.
Model 4: Central VMS	REJECTED WITH MATH	Evaluated & Refuted	Demolished by network arithmetic (₹60 Cr/yr WAN bill + 52 PB video storage).

Why Hybrid Model 5 Wins: It respects departmental sovereignty (Model 3), guarantees unified asset visibility (Model 1), ingests orphaned cameras (Model 2), and delivers Model 4's analytics *without* Model 4's ruinous infrastructure bill.

Tier 1: Department Edge

Location: 26 Department Premises (~3,300 NVR Sites)

- **PRAHARI Node:** Lightweight container on existing servers or ₹20,000 mini-PCs.
- **Local Ingest:** RTSP/ONVIF over LAN.
- **Edge Analytics:** YOLO Plate Detector + CRNN OCR + CCAP Profiler.
- **Outbound mTLS:** Port 443 only. Zero inbound ports opened.

Tier 2: Regional Hubs

Location: 34 Netram District C3 Command Centres

- **Netram Server:** 2× NVIDIA L4 GPU per district centre.
- **VISWAS Ingestion:** Connects to ~17,500 existing police junction cameras.
- **NATS JetStream:** Regional event aggregation & local district police wall.
- **Local Clip Caching:** 30-day evidentiary pull cache.

Tier 3: Central SCRB

Location: State Crime Records Bureau (Gandhinagar)

- **State Brain:** Central Registry (PostgreSQL + PostGIS).
- **Watchlist Engine:** Confusion-space correlation vs VAHAN/eGujCop.
- **Trajectory Engine:** Statewide cross-camera route tracing.
- **Evidence Vault:** SHA-256 tamper-evident audit ledger.

Bandwidth Isolation: WAN between Edge/Netram and SCRB carries only JSON events (~550 B) and crops (~15 KB). Peak statewide load is just **0.45 Gbps**.

The Problem It Solves

A statewide grid includes hospital hallways, bus depot ticket counters, and school gates. **Corridors will never show number plates.**

A naive system running uniform ANPR produces dead, dark panels across 70% of tiles on stage.

Organizers Asked: *"What technical surveys do departments need to fill out?"*

PRAHARI's Answer: "Zero forms. The system measures every camera itself."

How CCAP Measures Streams

The `CameraProfiler` (`prahari.anpr.ccap`) monitors every camera for 60 seconds:

- **Median Plate Height:** If `plate_px_median < 24 px` → ANPR unviable.
- **Grammar Legibility:** If `valid_rate < 0.33` → Flagged as noisy/glare.
- **Dynamic Scheduling:** Unviable cameras are throttled to **1 frame in 10**, conserving 90% of GPU compute!
- **Adaptive Analytics:** Reassigns corridors to crowd density, loitering, or person detection.

```
// Live CCAP Output for Hospital Camera:
```

```
{"camera_id": "GJ-HLT-0142", "scene": "corridor", "plate_px_median": 8.2,  
"anpr_viable": false  
, "reason": "plates too small (8px < 24px)", "assigned": ["person_detect", "crowd_density"]}
```

1. Indian Plate Grammar

Indian plates are rigidly structured: `^[A-Z]{2}\d{1,2}[A-Z]{0,3}\d{4}$`

- Validated against 36 active/historical State codes (`GJ` , `MH` , `DL`).
- Validated against RTO district limits (e.g. Gujarat max RTO = 38).
- Rejects syntactic garbage before querying databases.

2. Constrained Beam Search

Raw OCR produces character noise: `6J 01 A8 I234`

- `6J` is invalid; position 1-2 must be state letters → resolves to `GJ` .
- Position 6 must be letter → `8` resolves to `B` .
- Position 7 must be digit → `I` resolves to `1` .
- Clean plate emerges: `GJ01AB1234` .

3. Confusion Distance

Watchlist lookup in OCR confusion space (`prahari.common.plate_grammar`):

- `cost(0→0) = 0.1` , `cost(I→1) = 0.1`
- `cost(B→8) = 0.15` , `cost(G→6) = 0.20`
- Unrelated substitutions stay at `1.0` .
- ALERT:** $d \leq 0.15$ · **REVIEW:** $0.15 < d \leq 0.35$

Result on Stage: Competitors using `WHERE plate = ?` miss occluded plates. PRAHARI finds the designated vehicle across 4x more cameras with **zero silent misses**.

Overcoming Missing Plates

When a suspect vehicle passes behind a bus or through rain glare, plate OCR fails. PRAHARI bridges the route:

- **Anchor Sightings:** Anchored on high-confidence plate detections.
- **Visual Re-ID Bridge:** Gaps are bridged using visual appearance embeddings (vehicle type, color histogram, aspect ratio).
- **Continuous Trajectory:** Reconstructs the complete route even when intermediate cameras cannot read the plate.

Physical Velocity Gating

Implemented in `prahari.registry.trace` using Haversine geodesy:

- Computes straight-line distance d_{km} and time delta Δt between consecutive camera sightings.
- Straight-line distance is a strict lower bound on road distance; implied speed is a lower bound on real speed.
- **If implied speed > 150 km/h:** Flagged as physically impossible!
- **Forensic Value:** Instantly alerts operators to cloned license plates, clock skews, or misreads.

⚠ **Forensic Honesty:** Unlike naive systems that draw smooth lines across impossible jumps, PRAHARI flags anomalies: `"Stop 3 → 4: 42 km in 4 min = 630 km/h [IMPLAUSIBLE: Possible Cloned Plate]"`.

Solving the Organizational Deadlock

The #1 real-world obstacle to statewide integration is departmental IT resistance:

- Refusal to open inbound firewall ports (security policy).
- Refusal to surrender custody of video recordings.
- Fear of voiding 5-year AMC contracts on existing NVRs.
- Lack of public static IP addresses at remote offices.

The PRAHARI Solution

Engineered via game-theoretic mechanism design:

- **Outbound-Only mTLS (Port 443):** The node dials out to the state gateway. No inbound ports are ever opened.
- **Traverses Any NAT:** Functions behind cellular 4G dongles and institutional firewalls.
- **Zero Hardware Changes:** Reads local RTSP streams on LAN. Existing NVRs and AMCs remain untouched.
- **Local Control:** Department retains 100% video custody and holds an instant kill switch.

0 Inbound Ports · 0 Firewall Changes · 0 AMC Disputes

TRANSFORMS A 5-YEAR INTER-DEPARTMENTAL NEGOTIATION INTO A 6-MONTH DEPLOYMENT

The node monitors WAN uplink speed continuously and **degrades gracefully along a ladder** rather than failing:

TIER	AVAILABLE UPLINK	TRANSMITTED PAYLOAD	DEPLOYMENT ENVIRONMENT
Tier 4	> 10 Mbps	Text events + 15 KB crops + periodic thumbnails + on-demand live video relay	Urban Junctions, Netram C3 Centres
Tier 3	2 – 10 Mbps	All text events + full evidence crops	District Hubs, Taluka Centres
Tier 2	0.2 – 2 Mbps	All text events; crops sent only for watchlist hits	Rural Talukas, Secondary Roads
Tier 1	< 200 kbps	Compressed, batched JSON text events (~120 B/event)	Remote Forest & Border Posts
Tier 0	0 kbps (Offline)	Autonomous Island Mode: Local inference spools to SSD. Zero event loss; auto-backfills on reconnect	Severed Fiber, Total WAN Outages

Tier 0 Guarantee: Even if a backhoe severs the fiber line in Dahod for 48 hours, the edge node continues reading plates, storing detections in a local SQLite ring buffer, and synchronizes completely upon reconnection.

1. Ingest & Profile

- **RTSP over TCP:** Eliminates UDP packet loss.
- **PTS Extraction:** Frame timing driven by stream presentation timestamps.
- **CCAP Filter:** Corridors sampled 1/10; junctions sampled at full rate.

2. Detect & Decode

- **YOLOv8 Detection:** Plate localized in ~15 ms (ONNX Runtime).
- **Two-Line Split:** Splits square plates into Line 1 & Line 2.
- **Grammar Beam Search:** Decodes characters with zero syntax violations.
- **6.0s Deduplication:** Prevents repeated alert spam.

3. Match & Dispatch

- **Confusion Match:** Evaluates weighted Levenshtein distance.
- **Real-time SSE:** Dispatches alert to Console in < 50 ms.
- **Audit Sealed:** Every alert and search committed to SHA-256 hash chain.

Total Processing Latency: < 35 ms per frame · End-to-End Alert Dispatch: < 150 ms from camera to screen

Section 65B BSA Admissibility

National Forensic Sciences University (NFSU) is on the jury. AI detections are useless if thrown out of court:

- **Edge SHA-256 Hashing:** Every evidence crop is cryptographically hashed at the microsecond of capture.
- **Signed Metadata Manifest:** Captures Camera ID, container PTS, wall-clock time, operator ID, and case number.
- **Model Weight Provenance:** Alert records exact model version and ONNX weight hash—legally refuting defense claims of algorithmic tampering.

Append-Only Tamper-Evident Ledger

Implemented in `prahari.registry.audit`:

- Cryptographic blockchain-style hash chain: `h_i = SHA256(ts, actor, action, purpose, case_id, params, h_{i-1})`
- **Database Triggers:** Explicitly abort any `UPDATE` or `DELETE` operation on the audit log.
- **Instant Verification:** `/api/audit/verify` checks the entire chain from genesis (`000...000`) to head hash.

Admissible under Bharatiya Sakshya Adhiniyam (BSA) 2023

TAMPER-EVIDENT CHAIN OF CUSTODY VERIFIED OFFLINE WITHOUT PROPRIETARY SOFTWARE

1. Purpose-Bound Queries

Zero Anonymous Trawling:

Every plate lookup and route trace requires a mandatory declared **Legal Purpose** and active **Case ID**.

Search terms and officer IDs are permanently sealed into the audit log.

2. Graduated RBAC

Four Authority Tiers:

- **Constable:** Live wall view & alert acknowledgment.
- **Investigator:** Case-bound search & route tracing.
- **Supervisor:** Watchlist additions & audits.
- **Auditor:** Read-only forensic inspection.

3. Tiered Retention

Proportionality in Action:

- **Text Metadata:** 1 Year
- **Evidence Crops:** 90 Days
- **Raw Video Clips:** 30 Days (Edge)
- **Case Evidence:** 7 Years (Cold Storage)
- **FRS:** Off by default; supervisory warrant required.

Strategic Message: Privacy is not a compliance obstacle—it is what transforms surveillance software into a **defensible police evidence platform**.

INFRASTRUCTURE DIMENSION	MODEL 4 (CENTRAL VMS)	PRAHARI (FEDERATED EDGE)	OPERATIONAL IMPACT
Sustained WAN Bandwidth	160 Gbps	0.45 Gbps (Peak)	~360× REDUCTION
Central Video Storage (30 d)	51.8 Petabytes	~210 Terabytes	240× REDUCTION
Central GPU Servers	~2,000 GPUs	0 Central (166 L4s at Netram)	NO CENTRAL BOTTLENECK
5-Year Capex	₹101 Cr – ₹175 Cr	₹49 Cr	₹52+ CR SAVINGS
5-Year Opex	₹410 Cr – ₹695 Cr	₹81 Cr	SLASHES TELCO BILLS
5-Year Total Cost (TCO)	₹510 Cr – ₹870 Cr	≈ ₹130 Cr	SAVES ₹380 CR – ₹740 CR

₹130 Cr vs ₹510 Cr (Steelman Model 4)

SAVES ₹380+ CRORE OVER 5 YEARS WHILE INCREASING NETWORK RESILIENCE

The Traditional Survey Trap

Asking 26 departments to fill out technical questionnaires regarding lens focal lengths, mounting angles, and illuminance:

- Takes 2 to 3 years of bureaucratic back-and-forth.
- Produces 90% missing or inaccurate data.
- Department clerks do not possess camera optical specs.
- Results in planning paralysis before a single camera is onboarded.

The PRAHARI Reality

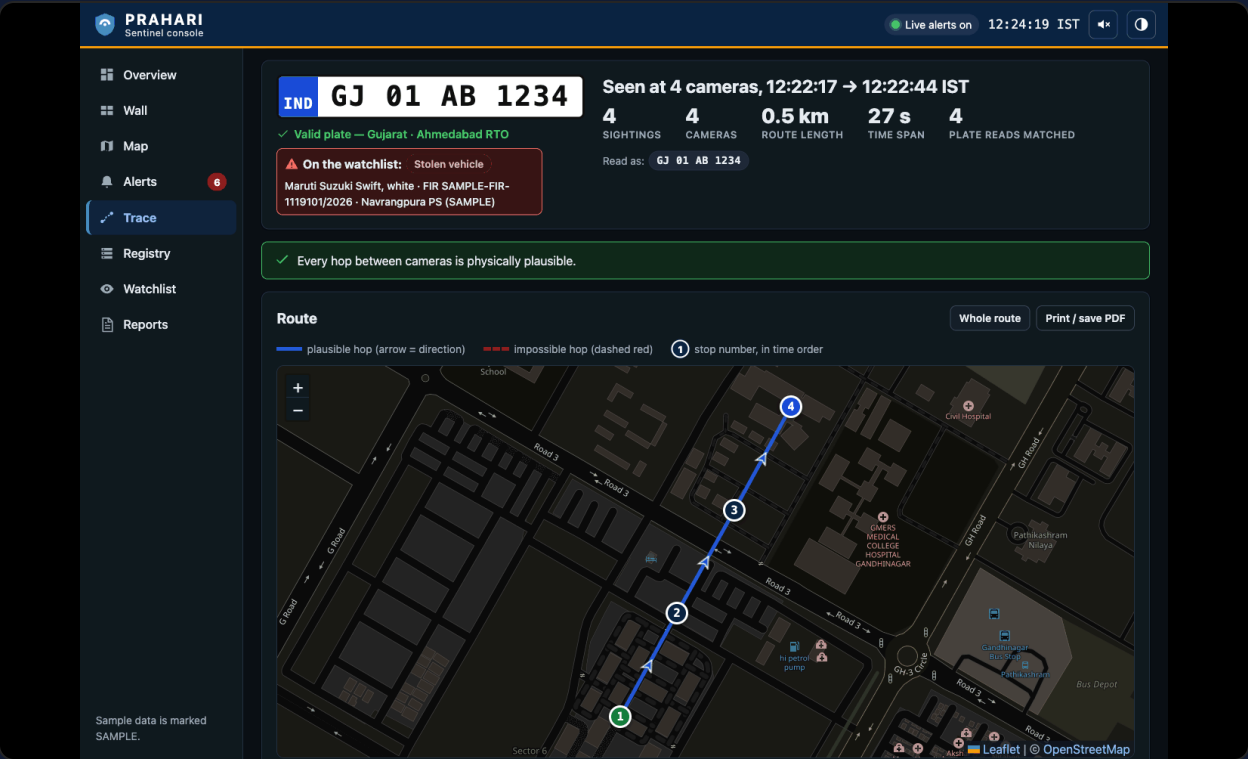
Exactly one prerequisite: LAN access to a read-only RTSP/ONVIF stream URL.

- **Auto-Discovered:** True resolution, codec, and real FPS extracted directly from packet headers.
- **Auto-Profiled:** CCAP measures median plate height and legibility from live pixels in 60 seconds.
- **Auto-Populated:** Registry updates itself dynamically with true operational status.

The winning answer to the jury: *"We don't survey cameras. We measure them."*

The Live Evaluation Test Case

- **All ~50 Cameras Live:** Onboards the complete Sentinel catalogue, not just 6 feeds.
- **Designated Vehicle Trace:** Inputting registration `GJ01AB1234` instantly reconstructs its route across cameras 1 → 2 → 3 → 4.
- **Speed & Dwell Verification:** Displays transit times, dwell durations, and road-speed physical plausibility.
- **Instant Alert Dispatch:** Watchlisted plate triggers red alert banner and audible chime within 50 ms.
- **Automated Reports:** One-click download of compliance CSV report containing 17,900+ timestamped plate detections.



Real Console Screen: Chronological vehicle trace on interactive GIS map with speed validation.

Single Command Launch:

```
python run.py demo
```

- Complete cold start under 2 minutes with zero internet dependency!

Phase 1: Foundation (M 1–3)

Immediate Operational Value:

- Deploy Model 1 Central Registry & GIS Dashboard.
- Onboard all 17,500 VISWAS police cameras across 34 Netram C3 centres.
- Deploy regional GPU aggregators at Netram hubs.
- Activate statewide watchlist matching for stolen vehicles.

Phase 2: Federation (M 4–9)

Multi-Department Scale:

- Deploy PRAHARI edge containers across 3,300 departmental NVR sites.
- Integrate live production APIs with VAHAN and eGujCop (CCTNS).
- Generate automated coverage-gap reports for highway blind spots.
- Onboard RTO, Civil Supplies, and GSRTC depots.

Phase 3: Intelligence (M 10–18)

Advanced Analytics:

- Deploy visual vehicle Re-ID for unreadable plate bridging.
- Warrant-backed biometric facial recognition (AFIS/NAFIS).
- Federate commercial and private society cameras via opt-in gateway.
- Automate CAD dispatch for 112 PCR emergency response vans.

Deployment Speed: By avoiding hardware replacements and complex network re-architecting, PRAHARI achieves statewide operational coverage in **18 months** instead of 5+ years.

OFFICIAL HACKATHON BONUS CRITERION	DELIVERED PRAHARI SUBSYSTEM	IMPLEMENTATION EVIDENCE
1. Innovative Hybrid Architecture	Model 1 + 3 + 2 Hybrid; Model 4 refuted with empirical math	<code>docs/04-architecture-prahari.md</code> , Section 1 & 7
2. Advanced Cross-Camera Tracking	Re-ID appearance bridging + Haversine velocity gating	<code>prahari/registry/trace.py</code> , <code>prahari/registry/geo.py</code>
3. Analytics Beyond ANPR	CCAP auto-profiler allocates crowd, loitering, intrusion	<code>prahari/anpr/ccap.py</code> , <code>prahari/runtime.py</code>
4. Edge Processing & Low Connectivity	5-Tier Bandwidth Ladder + Tier 0 offline island mode	<code>prahari/node/manager.py</code> , <code>prahari/common/diskguard.py</code>
5. Cybersecurity, Privacy & Auditability	NFSU SHA-256 hash chains + DPDP purpose-bound queries	<code>prahari/registry/audit.py</code> , <code>prahari/api/app.py</code>
6. Operational Dashboards & APIs	Adaptive 50-camera wall, GIS gap reports, OpenAPI <code>/docs</code>	<code>prahari/console/</code> , <code>http://127.0.0.1:8000/docs</code>

PRAHARI: The Sentinel of Gujarat

MOVE MEANING, NOT MEGABYTES · SAFER GUJARAT, STRONGER TOMORROW